BITS Pilani

Pilani|Dubai|Goa|Hyderabad

Introduction to Data Science

Data Wrangling

- *The slides presented here are obtained from the authors of the books and from various other contributors. I hereby acknowledge all the contributors for their material and inputs.*
- *We have added and modified slides to suit the requirements of the course.*

Statistical Descriptions

Basic Statistical Descriptions of Data

- Motivation
  - To better understand the data: central tendency, variation and spread
- Data dispersion characteristics
  - median, max, min, quantiles, outliers, variance, etc.
- Numerical dimensions correspond to sorted intervals
  - Data dispersion: analyzed with multiple granularities of precision
  - Boxplot or quantile analysis on sorted intervals
- Dispersion analysis on computed measures
  - Folding measures into numerical dimensions
  - Boxplot or quantile analysis on the transformed cube

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Measuring the Central Tendency

- <u>Mean (algebraic measure) (sample vs. population):</u>

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \mu = \frac{\sum x}{N}$$

Note: *n* is sample size and *N* is population size.

- Weighted arithmetic mean:

$$\bar{x} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

- Trimmed mean: chopping extreme values

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Measuring the Central Tendency

- Median:
  - Middle value if odd number of values, or average of the middle two values otherwise
  - Estimated by interpolation (for *grouped data*):

$$median = L_1 + \left(\frac{n/2 - (\sum freq)_l}{freq_{median}}\right) width$$

	age	*frequency*
	1–5	200
	6–15	450
	16–20	300
Median interval →	21–50	1500
	51–80	700
	81–110	44

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Measuring the Central Tendency

- <u>Mode</u>
  - Value that occurs most frequently in the data
  - Unimodal, bimodal, trimodal
  - Empirical formula:

$$mean - mode = 3 \times (mean - median)$$

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Probability Distribution

Probability distributions help us model and quantify uncertainty and variability in data.

Probability distributions also help us to analyze data and draw conclusions by describing the likelihood of different outcomes or events.

A frequently used probability density function (pdf) is Normal or Gaussian function.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Measures Data Distribution: Variance and Standard Deviation

- Variance and standard deviation (*sample: s, population: σ)*
  - **Variance**: (algebraic, scalable computation)
    - Q: Can you compute it incrementally and efficiently?

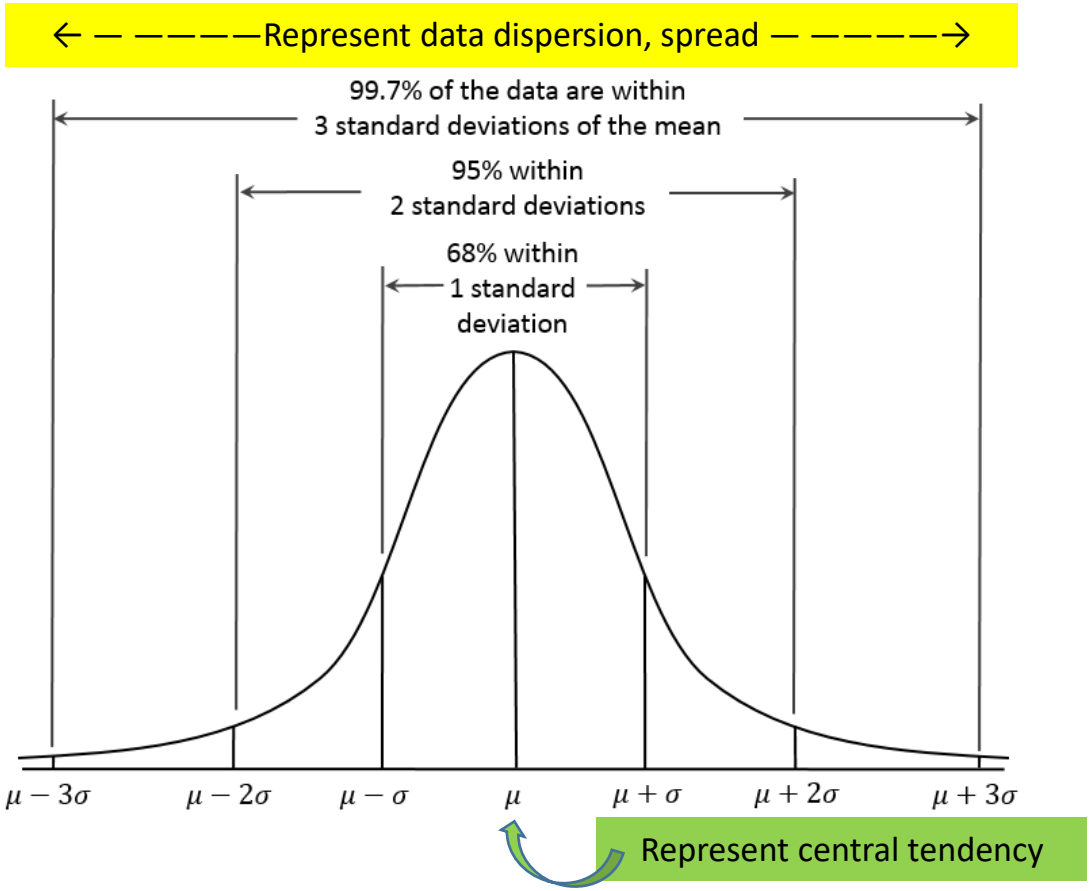
$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}[\sum_{i=1}^{n} x_i^{\,2} - \frac{1}{n}(\sum_{i=1}^{n} x_i)^2]$$

Note: The subtle difference of formulae for sample vs. population
- n : the size of the sample
- N : the size of the population

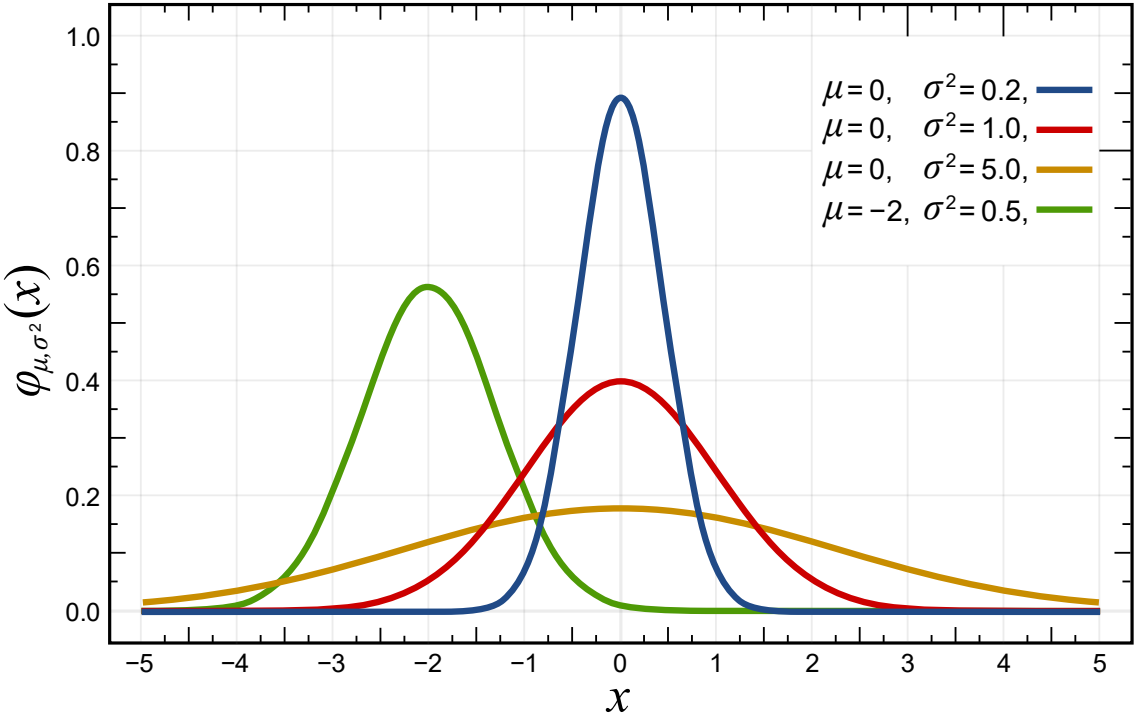
$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{n}(x_i - \mu)^2 = \frac{1}{N}\sum_{i=1}^{n} x_i^{\,2} - \mu^2$$

- **Standard deviation** *s (or σ)* is the square root of variance s^2 *(or* σ^2*)*

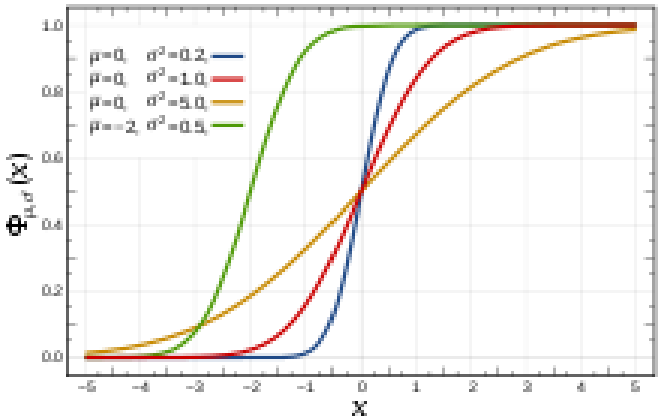
Properties of Normal Distribution Curve



Properties of Normal Distribution Curve



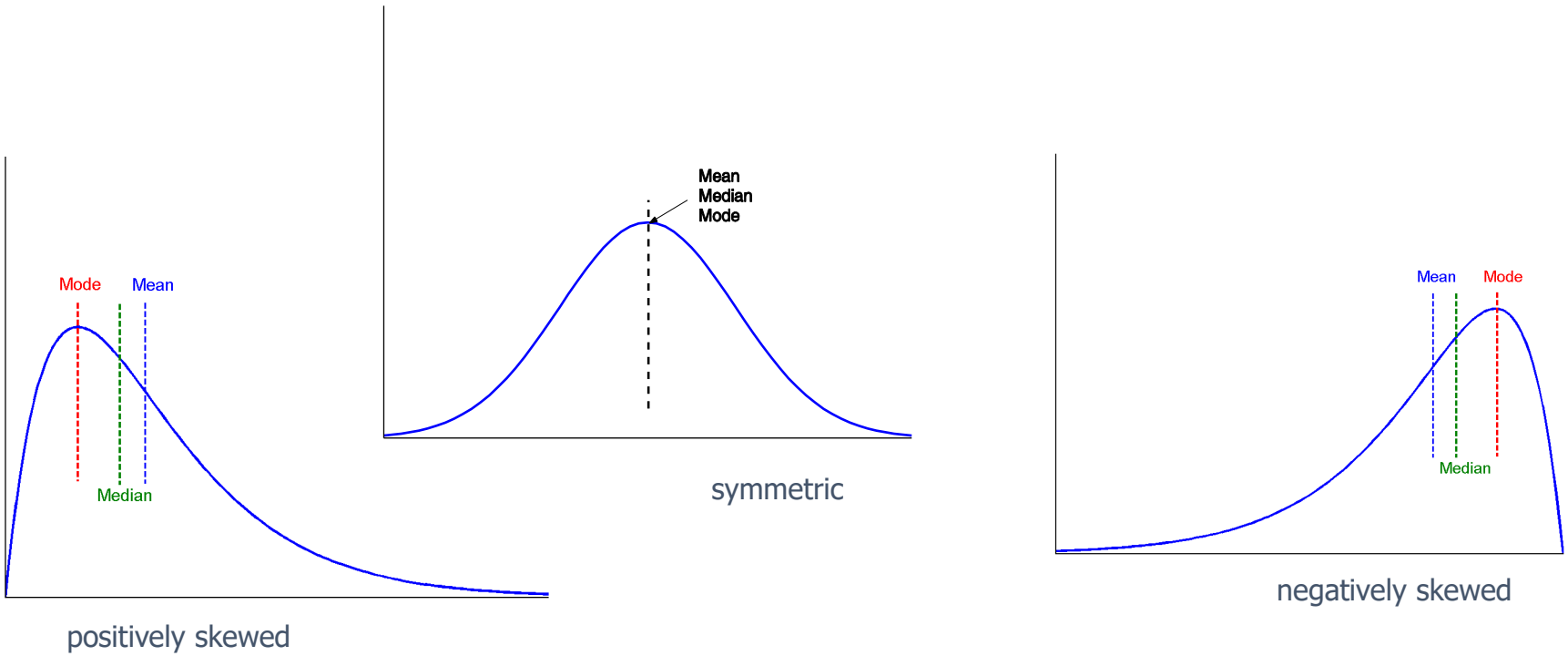
Probability density function



Cumulative distribution function

Symmetric vs. Skewed Data

- Median, mean and mode of symmetric, positively and negatively skewed data



Skewness and Kurtosis

Skewness is a measure of symmetry (more precisely, the lack of symmetry).

For univariate data $Y_1, Y_2, \ldots, Y_N$, the formula for skewness is:

$$g_1 = \frac{\sum_{i=1}^{N} (Y_i - \bar{Y})^3 / N}{s^3}$$

where $\bar{Y}$ is the mean, s is the standard deviation, and N is the number of data points.

The above formula for skewness is referred to as the Fisher-Pearson coefficient of skewness

https://www.itl.nist.gov/div898/handbook/eda/section3/eda35b.htm

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Skewness and Kurtosis

The skewness for a normal distribution is zero, and any symmetric data should have a skewness near zero.

- Negative values for the skewness indicate data that are skewed left (long tail to the left) and
- Positive values for the skewness indicate data that are skewed right (long tail to the right)

https://www.itl.nist.gov/div898/handbook/eda/section3/eda35b.htm

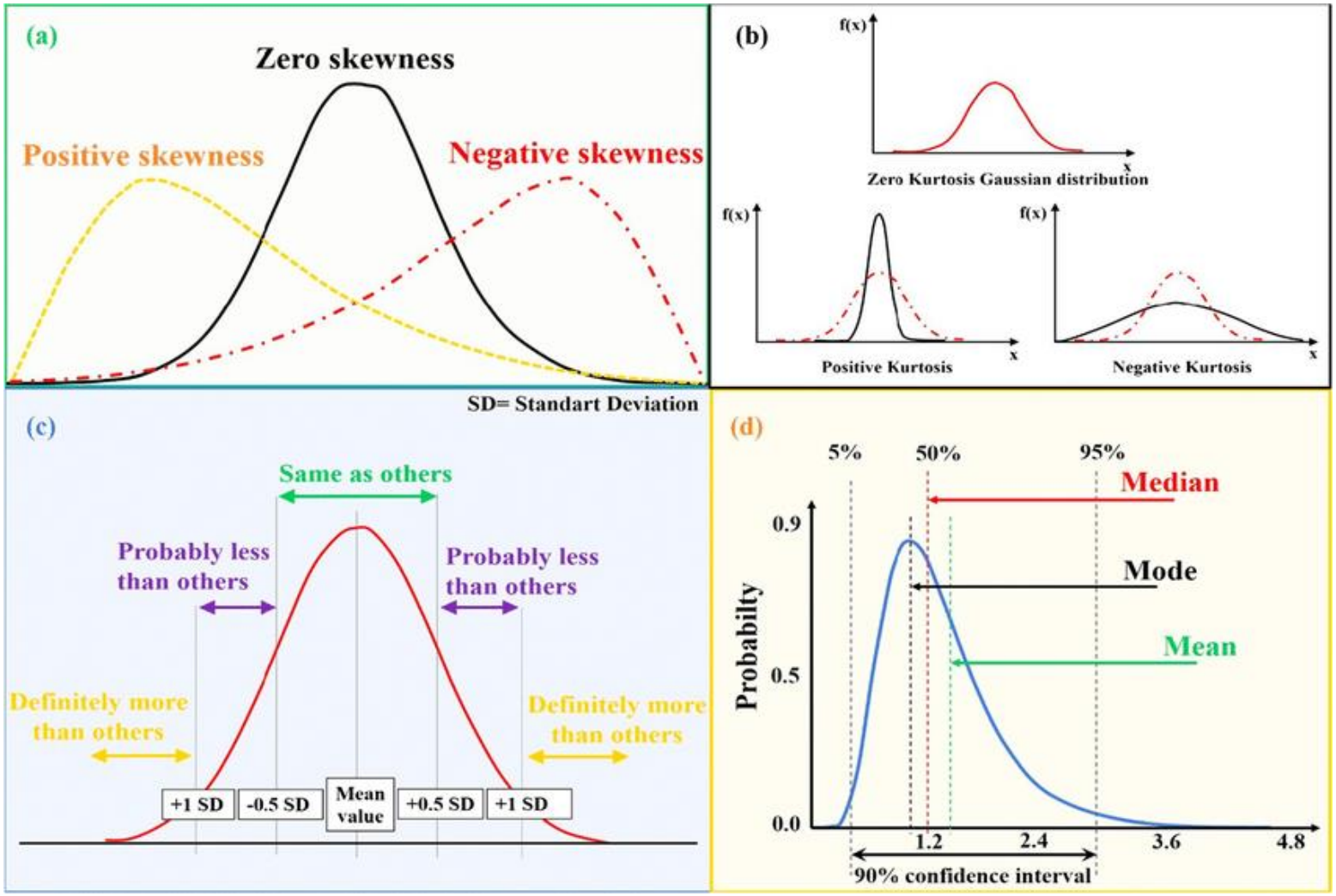
Skewness and Kurtosis

Kurtosis is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution.

- The data sets with high kurtosis tend to have heavy tails, or outliers.
- Data sets with low kurtosis tend to have light tails, or lack of outliers.

https://www.itl.nist.gov/div898/handbook/eda/section3/eda35b.htm

Skewness and Kurtosis



https://www.researchgate.net/figure/Statistical-moments-such-as-a-skewness-b-kurtosis-c-variance-and-d-mean_fig4_353016479

Measuring the Dispersion of Data

- Quartiles, outliers and boxplots
  - **Quartiles**: Q_1 (25th percentile), Q_3 (75th percentile)
  - **Inter-quartile range**: IQR = $Q_3 - Q_1$
  - **Five number summary**: min, Q_1, median, Q_3, max
  - **Boxplot**: ends of the box are the quartiles; median is marked; add whiskers, and plot outliers individually
  - **Outlier**: usually, a value higher/lower than 1.5 x IQR (on both sides of box from Q1 to Q3)
- Variance and standard deviation (*sample: s, population: σ)*
  - **Variance**: (algebraic, scalable computation)

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}[\sum_{i=1}^{n} x_i^{\,2} - \frac{1}{n}(\sum_{i=1}^{n} x_i)^2]$$

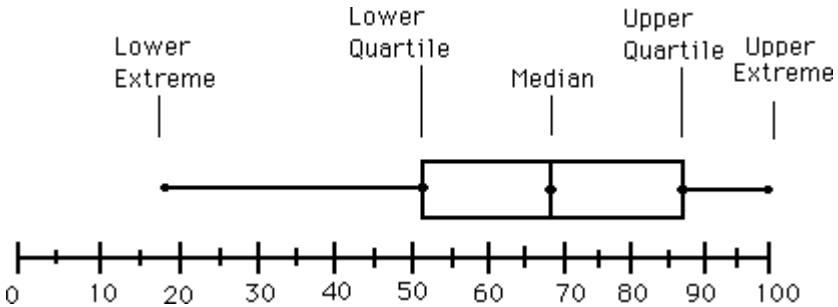
$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{n}(x_i - \mu)^2 = \frac{1}{N}\sum_{i=1}^{n} x_i^{\,2} - \mu^2$$

  - **Standard deviation** *s (or σ)* is the square root of variance s^2 *(or* σ^2*)*

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Boxplot Analysis

- **Five-number summary** of a distribution
  - Minimum, Q1, Median, Q3, Maximum



- **Boxplot**
  - Data is represented with a box
  - The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
  - The median is marked by a line within the box
  - Whiskers: two lines outside the box extended to Minimum and Maximum
  - Outliers: points beyond a specified outlier threshold, plotted individually

Example

Following is an ordered list of observations of a variable. Compute 5 point summary.

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Solution:

Min: 13

Q1: 20

Median: 25

Q3: 35

Max: 70

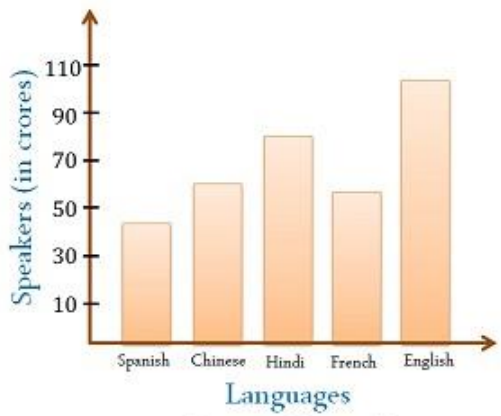
Any possible outliers here?

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

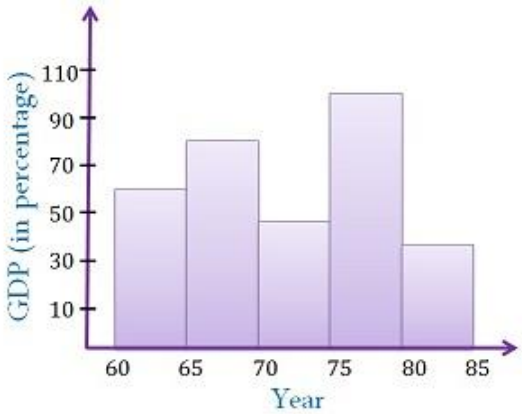
Graphic Displays of Basic Statistical Descriptions

- **Boxplot**: graphic display of five-number summary
- **Histogram**: x-axis are values, y-axis repres. frequencies
- **Quantile plot**: each value x_i is paired with f_i indicating that approximately 100 f_i % of data are $\leq x_i$
- **Quantile-quantile (q-q) plot**: graphs the quantiles of one univariant distribution against the corresponding quantiles of another
- **Scatter plot**: each pair of values is a pair of coordinates and plotted as points in the plane

Histogram Analysis



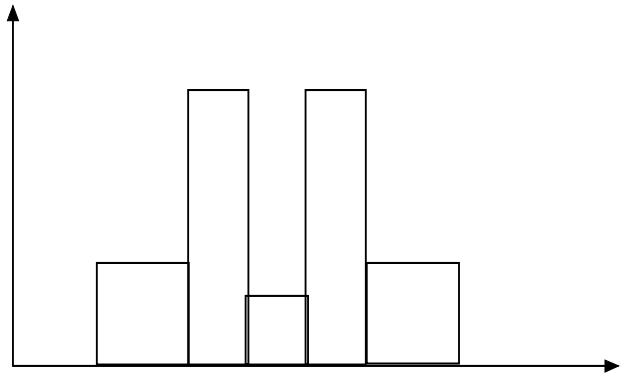
Bar Graph

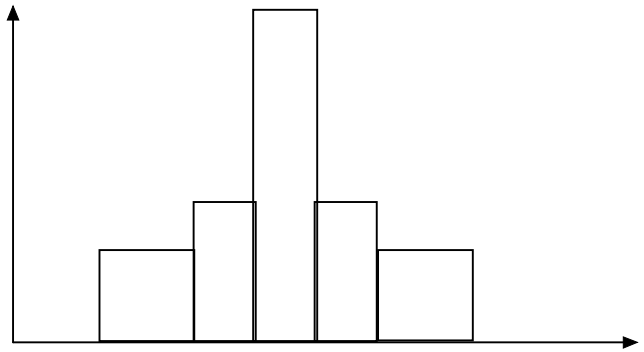


Histogram

- Histogram: Graph display of tabulated frequencies, shown as bars
- It shows what proportion of cases fall into each of several categories
- Differs from a bar chart in that it is the *area* of the bar that denotes the value, not the height as in bar charts, a crucial distinction when the categories are not of uniform width
- The categories are usually specified as non-overlapping intervals of some variable. The categories (bars) must be adjacent

Histograms Often Tell More than Boxplots

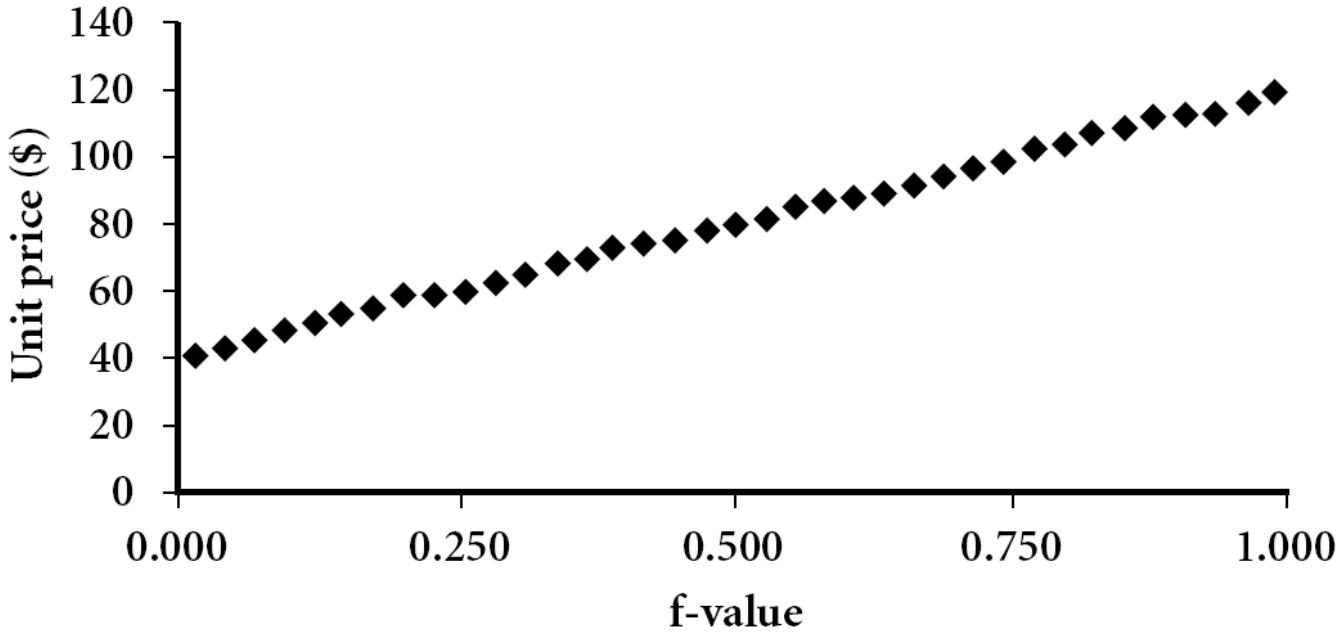




- The two histograms shown in the left may have the same boxplot representation
  - The same values for: min, Q1, median, Q3, max
- But they have rather different data distributions

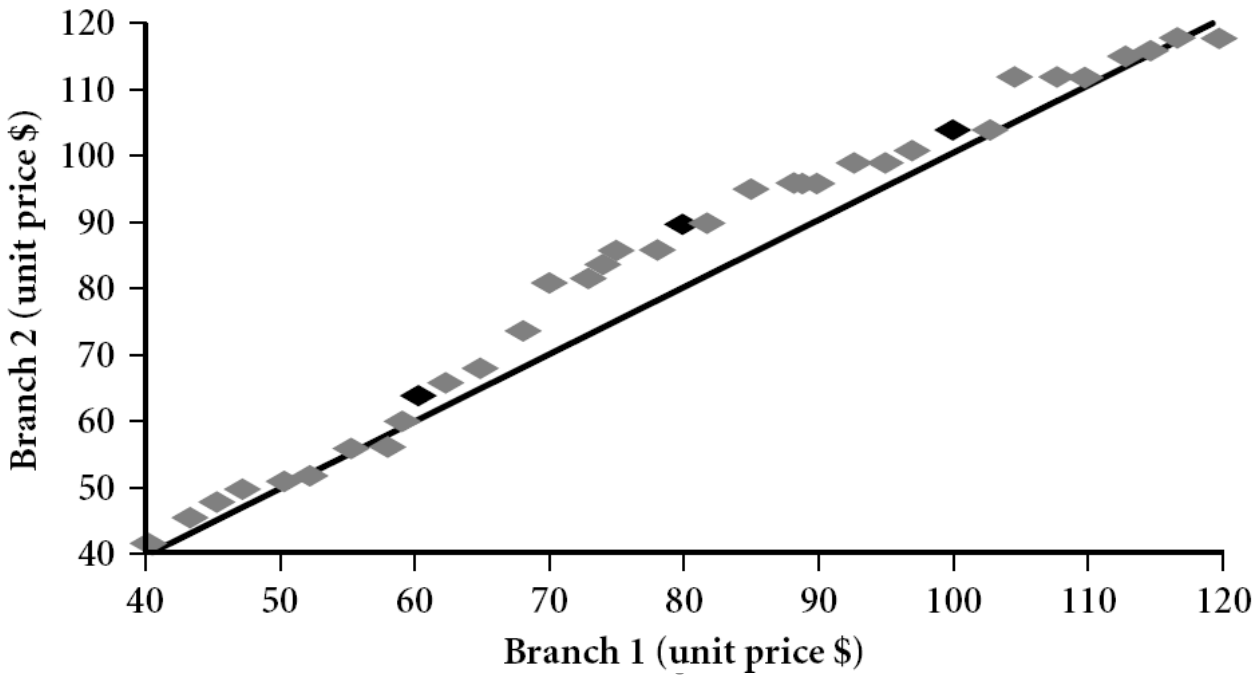
Quantile Plot

- Displays all of the data (allowing the user to assess both the overall behavior and unusual occurrences)
- Plots **quantile** information
  - For a data x_i data sorted in increasing order, f_i indicates that approximately 100 f_i% of the data are below or equal to the value x_i



Quantile-Quantile (Q-Q) Plot

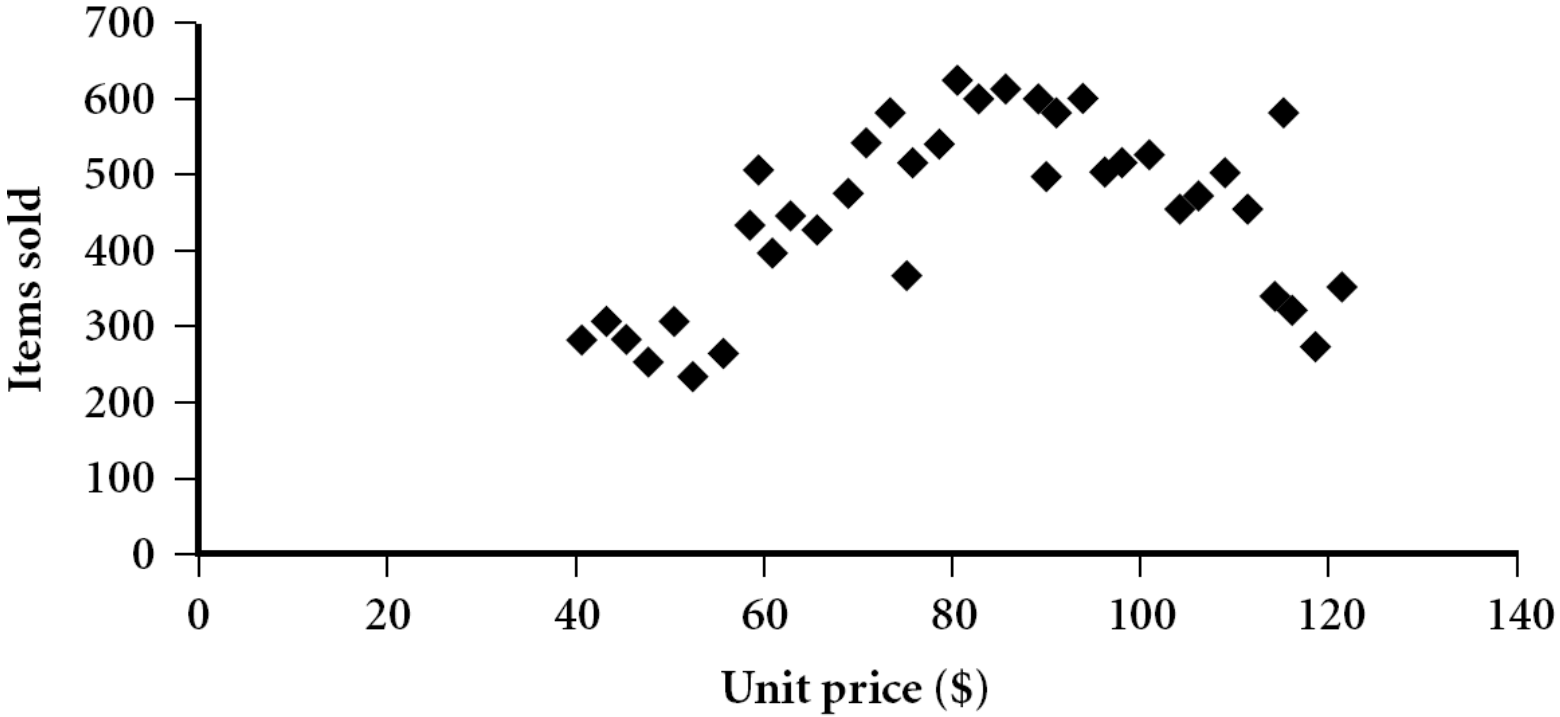
- Graphs the quantiles of one univariate distribution against the corresponding quantiles of another
- View: Is there a shift in going from one distribution to another?
- Example shows unit price of items sold at Branch 1 vs. Branch 2 for each quantile. Unit prices of items sold at Branch 1 tend to be lower than those at Branch 2.



BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

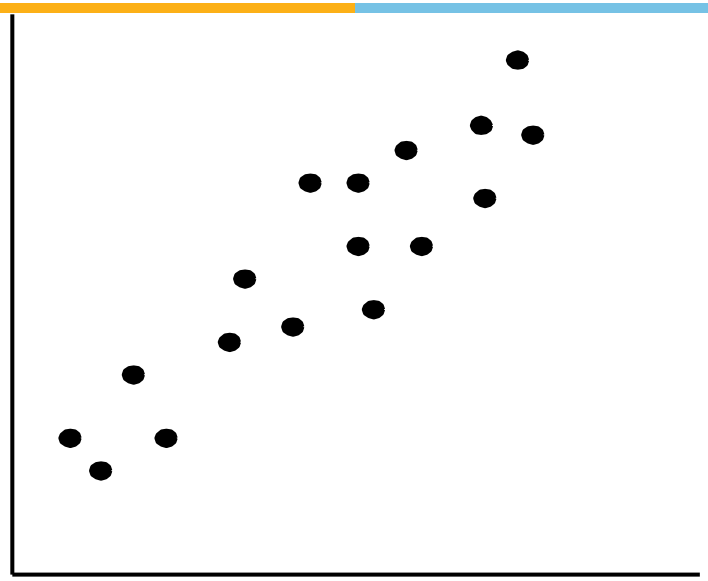
Scatter plot

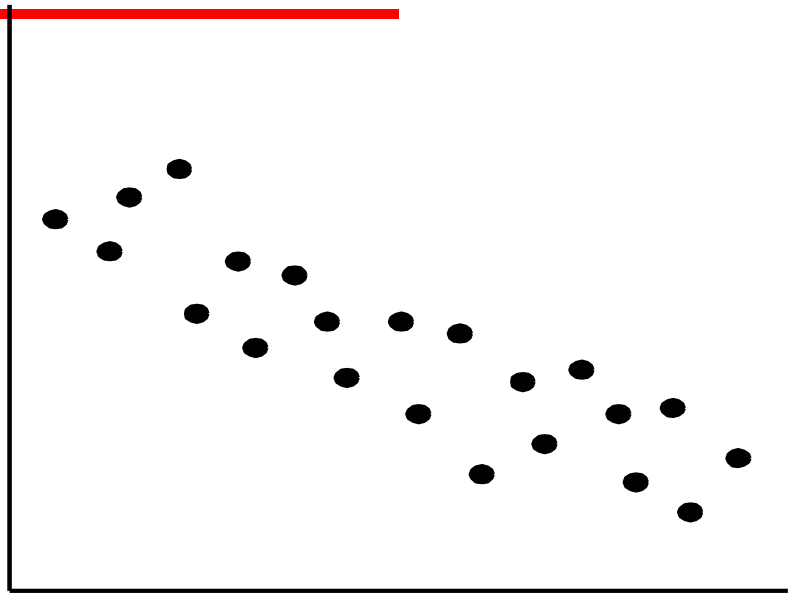
- Provides a first look at bivariate data to see clusters of points, outliers, etc
- Each pair of values is treated as a pair of coordinates and

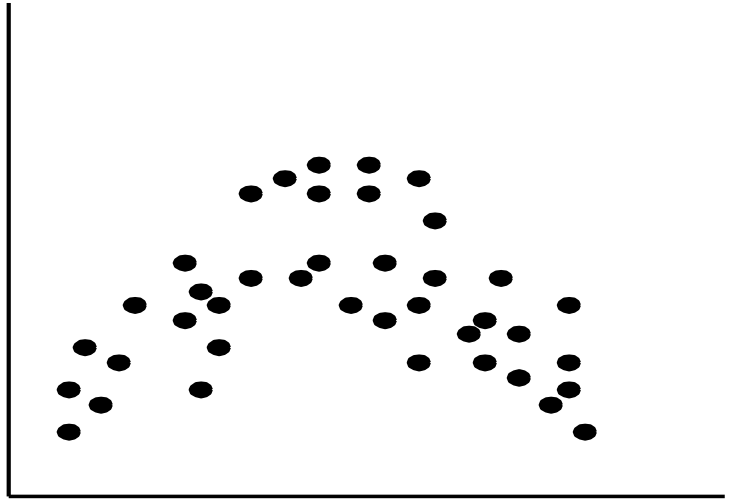


BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Positively and Negatively Correlated Data

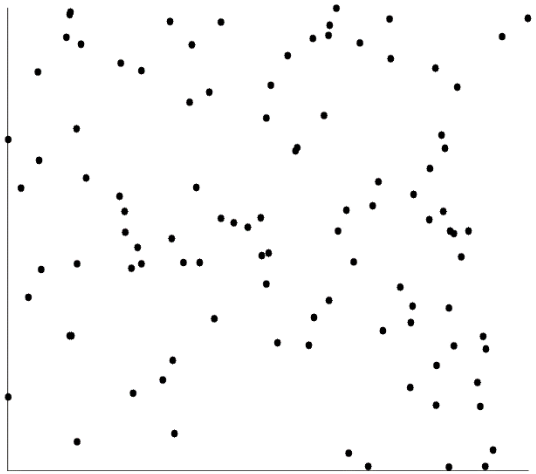


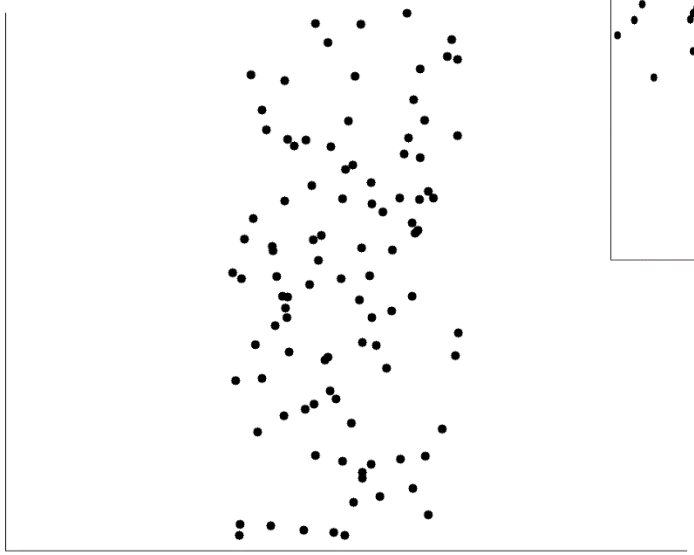


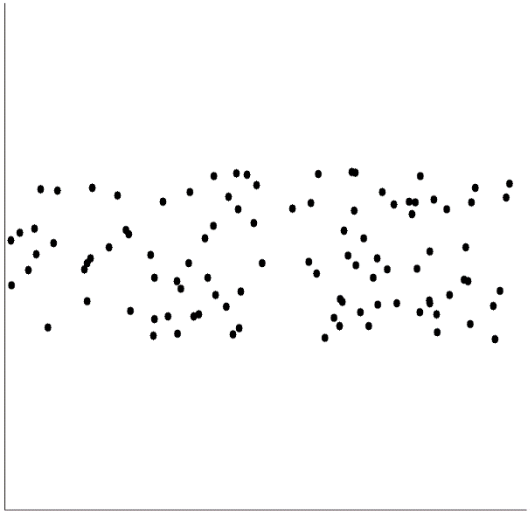


- The left half fragment is positively correlated
- The right half is negative correlated

Uncorrelated Data







Text Books

T1	Introduction to Data Mining, by Tan, Steinbach and Vipin Kumar
T2	Introducing Data Science by Cielen, Meysman and Ali
T3	Storytelling with Data, A data visualization guide for business professionals, by Cole Nussbaumer Knaflic; Wiley
T4	Data Mining: Concepts and Techniques, Third Edition by Jiawei Han and Micheline Kamber Morgan Kaufmann Publishers

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Thank You